

Nasan-Ochir Sukhbaatar

Applied AI Engineer / Tech Lead / Full-Stack and Mobile

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SUMMARY

Applied AI engineer and team lead with 8+ years shipping software in Mongolia across government, banking and consumer products. I train and ship models for a low-resource language: my fine-tuned Mongolian voice model dubs full drama episodes with distinct, emotional native voices. I also build grounded LLM products end to end, from data pipelines to evaluation and paid production use. Before that I led mobile and full-stack teams on a national e-library, an 8,500-seat arena platform, bank apps and enterprise systems for Shunkhlai Group and GS25.

SELECTED AI WORK

- **Mongolian drama dubbing model (trained myself).** Built a ~92-hour, 95-speaker Mongolian speech corpus and fine-tuned a voice model on it, including emotion training (angry, sad, tender delivery). Built per-character voice casting that keeps one consistent voice per character across a series, and an automated QA gate that checks every line for correct words, voice, gender, pitch, timing and a model-based listening test that matches human review 87-93% of the time. Built the subtitle side too: it reads the source subtitles (official tracks, or burned-in text read from the video frames), rewrites them as natural Mongolian sized to each line's timing, and burns Mongolian subtitles into the final episode. **Watch:** nasanochir.github.io/mongolian-ai-dubbing
- **Grounded civic AI (Open Parliament, 3rd place).** Search and chat over parliament agendas, transcripts and 950 laws (15k articles), built only on official open government data. Missing data is shown as a gap, never filled in by the model. Facts are kept only when backed by a verbatim official quote. A validation rule threw out 345 of 550 wrong document links. The assistant re-answers under a stricter grounding rule when it fails to cite sources.
- **Paid LLM product (astro.mn).** 14 personalized long-form reading types and a chat that remembers the user. Facts are computed in code and locked into the prompt so the model can't invent numbers. Live with paid checkout.
- **Mongolian language tools.** Mongolian meeting transcription and AI meeting notes SaaS (ZeroTech). A spell checker with 96% top-1 suggestion accuracy against 72% for the market leader, on a held-out benchmark.
- **AI adoption.** Designed Shunkhlai Group's 4-module employee AI training program. Use multi-agent coding workflows day to day.

EXPERIENCE

ZeroTech LLC

Jul 2023 - Oct 2026

Senior Software Developer (backend, web, mobile)

GS25 Mongolia: mobile app and receipt admin panel Flutter, Next.js 16, React 19, Medusa UI, TanStack Query, Zustand, Docker

- Developed the GS25 convenience-store customer app and was the top contributor to the receipt administration panel used to manage and audit e-receipts.

Shunkhlai Group: Orange B2C app, B2B driver app, admin panel Flutter, Riverpod 3, Auto Route, Google Maps, Firebase, Codemagic CI/CD, Next.js 14, Mantine

- Lead developer on the Shunkhlai Orange consumer app (iOS/Android/web) for fuel and retail customers, with codegen-based architecture (Freezed, json_serializable, riverpod_generator).
- Built a B2B delivery-driver app with live maps and route handling, plus features in the Next.js operations panel.

Campus super-app with mini-apps Flutter, Riverpod 3, Firebase, Next.js, Vercel

- Architected and built nearly all of a campus super-app hosting pluggable mini-apps: events, facility requests, meeting-room booking, parking, indoor wayfinding map. Wrote the mini-app integration guide.

Car Parking platform (backend and admin) Node.js, Express + TSOA, Drizzle ORM, PostgreSQL, Docker, Jest

- Built ANPR plate-recognition entry/exit with Hikvision gate integration, eligibility and gate-direction rules, and entry-time pricing snapshots so later policy changes don't alter exit fees.
- Built QR payments: invoices, coupons, partial and double-payment handling, refunds, HMAC-verified callbacks. Also slot allocation, guest passes and an in-app notification center.

DOT / OnTap customer app Flutter, Riverpod, Dio, Firebase, I10n (MN/EN/KO)

- Top contributor (146 commits) to a trilingual customer app covering assets, finance and payments, branch and calendar booking, and push notifications.

APU apps React Native + Expo, tRPC, Drizzle ORM, PostgreSQL, Turbo monorepo

- Worked on the APU Route, APU DOT and APU Dairy apps.

goOrange: Microsoft Teams peer-recognition app (3rd place, Shunkhlai Hackathon) React 19, Vite, Supabase (Postgres, Edge Functions, RPC), Bot Framework, Microsoft Graph, MCP

- Built a Teams tab and bot where employees give each other "oranges": weekly allowances, reactions, rewards redemption, idea board. Includes Graph-synced directory, proactive notifications and an MCP/agent interface.

SteppeLink LLC (Steppe Link Holding)

Dec 2021 - Jul 2023

Team Lead, Mobile (team of about 5)

Esan: national digital library iOS, Android, Radium

- Project lead with a team of 8 on Mongolia's national e-library (ebooks, audiobooks, theses, newspapers, magazines) with offline reading.
- Integrated the Radium SDK for EPUB rendering, with pagination, highlights and reading-position sync.

Steppe Arena app

- Delivered the app for the 8,500-seat AIC Steppe Arena: event and sports ticketing, in-seat food ordering, ice-skating session booking, and plate-based smart-parking payment.

Odin Survey

- Built a survey and market-research platform (web, iOS, Android) where users earn points and coupons for completing surveys and business clients get aggregated insights.

Infinite Solutions LLC

Oct 2018 - Dec 2021

Senior Software Developer, Mobile

- Built and maintained mobile banking apps for major Mongolian banks: **Arig Bank, XacBank, Golomt Bank, Trade and Development Bank (TDB)**. Work covered secure login, accounts, transfers, card management and payments.
- Developed consumer apps **Shuffle** and **Elify**.
- Worked within a fintech vendor building payments, merchant and loyalty infrastructure under banking security and compliance requirements.

Astvision LLC

Apr 2018 - Oct 2018

Software Developer, Full-Stack

- Full-stack work on government e-service systems: early **e-Mongolia** platform work, the Ulaanbaatar city e-services portal (**e-service.ulaanbaatar.mn**), and **my.tog.mn**, the Ulaanbaatar Electricity Distribution Network customer portal for bills, consumption and service requests.

INDEPENDENT PRODUCTS AND AWARDS

- **3rd place, Open Parliament Hackathon (2026)**, organized by the Secretariat of the State Great Khural, The Asia Foundation and Unread Media. 24-hour build of an MP accountability platform (votes, attendance, speeches, bill compliance, grounded AI chat).
- **3rd place, Shunkhlai Hackathon** with goOrange, an employee recognition tool inside Microsoft Teams, later built out for production (see ZeroTech).
- **kinody.mn**: mobile-first Asian drama streaming for an audience that is 97% mobile, with QPay subscriptions and protected video delivery. Home of the dubbing model above.
- **gadaad.mn**: visa application helper for Mongolian travellers covering 10 countries. Passport scan autofill, document checklists with uploads, paid AI approval-chance scoring and arrival-card smart fill. Next.js 16, Supabase, Cloudflare Workers.
- **Koharu** (open source, MIT): contributor to a desktop manga image translator (Rust, Tauri 2, local LLMs) with text detection, OCR, inpainting and an MCP server.

EDUCATION

Sapienza University of Rome, M.Eng. program in Computer Science and AI (completed first year)

2017 - 2018

National University of Mongolia, Bachelor of Information Technology

2013 - 2017

SKILLS

Mobile	Flutter/Dart (Riverpod, Auto Route, Freezed), React Native/Expo, iOS/Android release, Firebase, Codemagic
Web	Next.js (App Router, 13-16), React 19, TypeScript, Tailwind, Mantine, TanStack Query, Zustand
Backend	Node.js (Express, TSOA, tRPC), Python (FastAPI), SQL (PostgreSQL), Supabase (RLS, Edge Functions), Drizzle, Redis, Docker
ML / Speech	Python (PyTorch, scikit-learn, pandas), model fine-tuning, speech corpus building, TTS, speech-to-text, evaluation and QA gates, GPU training
LLM Eng.	RAG and embeddings, grounding and fact validation, agents and MCP, streaming, prompt and cost optimization (Claude, OpenAI, Gemini)
Integrations	QPay, bank payment gateways, e-Mongolia XYP, Microsoft Graph/Teams, Hikvision ANPR, Google Maps
Cloud	Vercel, Cloudflare Workers, Azure, GCP, Linux GPU servers (RunPod)
Languages	Mongolian (native), English (professional)